POKHARA UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
# SCHOOL OF ENGINEERING

| Exam | Final Internal Examination 2025 Fall | | |
|---|---|---|---|
| Level | BE | FM | 100 |
| Program | Computer | PM | 45 |
| Year/ Part | IV/I | Time | 3 Hrs |

**Subject: Data Science and Analytics**

*Candidates are required to give answers in their own words as far as practicable.*
*The figure in the margin indicates full marks. Assume suitable data if necessary.*
***Attempt all the questions.***

| | | | |
|---|---|---|---|
| 1. | a) | How would you define data science? Explain the KDD process in brief. | 7 |
| | b) | A numeric variable has the following values.<br>46.93, 138.07, 64.77, 51.46, 15.38, 15.37, 11.32, 223.21, 120.23, 19.85, 19.80, 12.39, 55.79, 86.31, 3.99, 101.51, 30.71, 94.28, 102.39, 18.04<br>Standardize the values. | 8 |
| 2. | a) | Based on a survey, the variable x takes the following values.<br>0.76, 1.53, 0.12, 2.44, 2.66, 3.84, 1.03, 6.31, 2.03, 4.61, 9.53, 10.39, 3.69, 6.63, 6.4, 5.01, 2.99, 1.18, 0.35, 0.44, 1.99, 7.04, 3.05, 7.89, 1.44, 1.11, 7.53, 10.4, 1.72, 4.96.<br>Do you think x follows a normal or an exponential type distribution? Support your answer with appropriate plots. | 8 |
| | b) | Differentiate discrete and continuous numeric variables with examples.<br>What are nominal and ordinal categorical variables? Explain with examples | 7 |
| 3. | a) | Explain the importance of Pearson's correlation coefficient .<br>Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:<br>X = 10, 13, 15, 17, 19 and Y = 5, 10, 15, 20, 25. | 7 |
| | b) | The following table presents a hypothetical cross-tabulation between 'food preferences' and 'gender'.<br>**Food Preference** / **Female** / **Male** / **Other**<br>Likes buff momo: 8, 75, 1<br>Likes chicken momo: 60, 3, 0<br>Likes veg momo: 23, 19, 3<br>Do you think food preferences are correlated with gender? Justify using chi squared test with significance level 0.05. The critical value for the Chi-Squared statistic with 4 degrees of freedom at a significance level of 0.05 is 9.4877. | 8 |
| 4. | a) | What is multicollinearity in regression analysis?<br>How do variance-inflation factors (VIF) help detect it, and what steps can be taken if high multicollinearity is found? | 7 |
| | b) | Compare and contrast Principal Component Analysis (PCA) and K-means clustering, explaining their mathematical foundations and practical applications. | 8 |

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|---|---|---|---|
| 5. | a) | You are given the following data. Assuming that there is an unknown functional relationship y = f(x), estimate the value of y at x = 2.9 using the Nadaraya-Watson regression estimator. Take bandwidth h = 0.5. | 8 |

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $y$ | -2.3 | 0.5 | 9.3 | 20.8 |

b) What is the role of R-squared and Adjusted R-squared?
Compute R-squared for a regression model to measure how well it explains the variation in the dependent variable Y. **7**

| Hours Studies (X) | Exam Score (Y) |
|---|---|
| 1 | 50 |
| 2 | 60 |
| 3 | 70 |
| 4 | 80 |

**6.** **a)** Explain the concepts of Decision Trees and Logistic Regression in machine learning. **7**

**b)** Explain about autocorrelation and stationarity.
The following time series represents yearly production data: **8**

| t | X_t |
|---|---|
| 1 | 4 |
| 2 | 6 |
| 3 | 8 |
| 4 | 10 |
| 5 | 12 |
| 6 | 14 |

(a) Compute autocorrelation at lag $k=1$
(b) Divide the data into two equal parts and compute the mean
(c) Comment on stationarity

**7.** **Write short notes on:** *(Any two)* **2×5**

**a)** OLS estimation

**b)** Causation vs. correlation

**c)** Zipf's law

****** Best of Luck ******